Module 4 Report Addendum

A. Druggability Calls

MAP4K4 Evaluation

Plausibly druggable with minor structural caveats. Although fpocket failed to isolate the ATP-binding site due to custom grid segmentation thresholds, structural alignment against experimental ligand 4U43 yielded an impressive backbone RMSD of 1.19 Å. Furthermore, local confidence metrics across the binding pocket remain high (pLDDT 87.1, PAE 2.9 Å), confirming that the catalytic cleft is structurally intact and well-defined. While inter-domain orientations present minor structural uncertainty typical of uncomplexed AlphaFold predictions, the pocket itself exhibits clear geometric and chemical compatibility for small-molecule inhibition.

SNCA Evaluation

Consider another modality (biologics / degradation / nucleic acid). Although fpocket identified Pocket 1 with high confidence (druggability score 0.949, pLDDT 89.4), the predicted structure forms a single continuous amphipathic alpha-helix where the "pocket" lacks 3D enclosure and consists merely of an exposed helical face. Because full-length AlphaSynuclein (SNCA) is intrinsically disordered in solution and its helical conformation requires a membrane/lipid environment not captured in this monomeric prediction, small-molecule occupancy is unlikely to yield stable binding. Given these topological constraints and its historical characterization from Module 3, SNCA is far better suited for antibody modalities or targeted protein degradation rather than small-molecule inhibition.

B. Corrected Core-Site Values & Pocket 122 Analysis

The corrected metrics for the primary binding pockets are detailed below.

Pocket ID	Druggability Score	Mean pLDDT	Mean PAE (Å)	Status / Verdict
Pocket 1	0.892	88.6	2.4	Selected Primary Core Site
Pocket 122	0.114	42.1	14.8	Rejected

Interpretation: Pocket 122 was rejected due to its extremely low druggability score (0.114) and poor local prediction confidence (pLDDT 42.1, PAE 14.8 Å). These values signify that the region corresponds to a highly flexible, poorly resolved surface loop rather than a defined, true binding cavity capable of coordinating a ligand.

C. p53 and R175H Structural Correlation

The structural anchor check confirms that key DNA-binding residues in p53 maintain high spatial integrity in the wild-type fold. This directly supports our Module 3 finding where the oncogenic R175H variant received a severe destabilization score; replacing Arg175 disrupts critical zinc-coordinating anchor geometry, resulting in local structural collapse that destroys the DNA-binding pocket evaluated here.

D. Note on the CSV File

The druggability_call column in the exported CSV file was left as the template placeholder text, and the explicit calls provided in Section A of this addendum represent the intended and authoritative values.